

Ritvik Sharma

New York, NY

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Education

Columbia University

Aug 2025 - Dec 2026

MS in Computer Science - AIML & NLP Track

SRM University – Rank: 2/700+

Jun 2019 - May 2023

BTech in Computer Science - AI & ML Track

GPA: 9.86/10

Work Experience

Barclays, UK

Jul 2023 - Jul 2025

Software Engineer

Pune, India

- **Saved \$200,000 Annually:** Innovated utilities ensuring optimal AWS Lambda Usage. **Collaborated** cross-functionally to drive adoption. Awarded a **Barclays-Wide Recognition**.
- **Achieved 96% Performance Improvement** in Glue Jobs pertaining to CRUD operations in AWS DynamoDB dealing with 5M+ Records (**12 hours to 27 minutes**). **Led** the solution design through Spark, Dynamic Frames, and Batch Write, **coordinating with platform teams** for seamless implementation.
- **Reduced error-resolution times by 75%** by designing and developing a complex error-routing utility via Python-based Lambdas, IBM Netcool, and SNS filters. Played a **key role in facilitating communication** between development and operations teams to align on system-wide logging strategies.
- **Executed No-Downtime Deployments:** Integrated Blue-Green-Canary strategies into the system, **advised leadership, aligned cross-domain teams**, and ensured uninterrupted user experience, earning Barclays-wide recognition.
- **Optimized CI-CD Pipelines with 63% higher success rates:** Fostered team collaboration across DevOps and application teams, earning organization-wide recognition at Barclays.
- **Spearheaded the engineering and development** of the integration layer connecting Onboarding, Rewards, Payments, and other modules for Barclays' first cloud-based credit card platform, **ArrowCard**.

Projects

VELVÉT - Luxury E-Commerce Website — AWS (SageMaker, OpenSearch), React, JavaScript, Python, NLP

[Link to Live Demo](#)

- Created full-scale real-world functionality, including **ML-based semantic product search, wishlist management, cart operations, dummy order placement, and similar product recommendations** for a seamless user experience.
- Implemented a secure and high quality AWS backend via IAM Roles, Basic DDoS Prevention and performed **Selenium-based unit + integration tests** to ensure Production-Grade Quality.

Hackathon Winner: Mercury — Python, Flutter, NLP, REST APIs, gRPC

[Link to Web/Video Demo](#)

- **Developed RESTful APIs** for an AI-powered chatbot leveraging ML and NLP to accurately extract food order details from natural language inputs such as, "The people want **2 Veg Margherita Pizzas** with **Olives** from **Dominos**".
- **Directed a team of 3 to architect and deliver** the backend of a mobile app using Python and Flutter, integrating ALBERT (LLM for NLP), Conditional Random Fields for probabilistic prediction, and Speech-to-Text capabilities.
- **Collaborated & orchestrated team victory** in a 24-hour hackathon by strategically allocating tasks based on **analyzing individual strengths**, demonstrating exceptional **teamwork and impactful communication**.

Technical Skills

Languages: Python, JavaScript, Java, C/C++, SQL (Postgres, MySQL), NoSQL, HTML/CSS

Frameworks: React, Angular, Node.js, Flask, JUnit, JQuery, FastAPI, Tomcat, Django, Selenium, Flutter

Developer Tools: Git, GitHub, Jenkins, Docker, Amazon Web Services (AWS), GitLab, Linux

Technologies: PyTorch, NLP, Pandas, NumPy, Scikit-Learn, Machine Learning, Computer Vision, OpenCV, matplotlib, Kubernetes, LLM

Extracurriculars

- Sharma, Ritvik (April 2025). "VectorQuest: Faster and Better Open-Domain Question-Answering System" **Published in Springer Nature Switzerland AG** for the innovation leading to **50 times Faster Prediction**.
- **Providing one-on-one tutoring** to Student Athletes at Columbia in quantitative and computational subjects, including **Probability, Machine Learning, Calculus, and Data Structures & Algorithms**.
- **Inter-University Hackathon Winner** for Building an AI-Powered Chatbot for ordering food. [Check Out the Demo Here](#).